

Complementary and Integrative Therapies for Lower Urinary Tract Diseases



Donna M. Raditic, DVM, CVA

KEYWORDS

- Complementary and alternative medicine • Integrative medicine
- Lower urinary tract disease (LUTD) • Urinary tract infection • Urolithiasis
- Urinary tract tumor • Supplement • Herbs

KEY POINTS

- There is a growing demand for the use of integrative medicine in veterinary medicine.
- Evidence-based research using integrative medicine in veterinary patients with lower urinary tract diseases is scarce.
- Translational research with animal models of human lower urinary tract disease (LUTD) is an opportunity to expand our knowledge of etiopathogenesis and identifying complementary treatments.
- Translational evidence-based research is needed to accelerate the use of integrative health care in both human and veterinary medicine where there is a concern for antimicrobial resistance.

In 2014, the National Center for Complementary and Alternative Medicine changed its name to the National Center for Complementary and Integrative Health (NCCIH). Complementary approaches have grown in use to where they are no longer considered an alternative to medical care. For example, more than half of Americans report using a dietary supplement and spend nearly \$4 billion annually on spinal manipulation therapy. Large population-based surveys have been done and found that the use of alternative medicine, which has been defined as unproven practices used in place of conventional medicine, is actually rare. The NCCIH defines integrative health care as the combination of complementary approaches into conventional treatment plans and has widespread use in human medicine. The goal of an integrative approach is to enhance overall health, prevent disease, and to alleviate debilitating symptoms, which often affect patients with chronic diseases and affect outcomes. The NCCIH states that despite the widespread use of integrative health care, the scientific evidence for many complementary approaches is still needed.¹

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Integrative Medicine, Nutrition Service, Department of Small Animal Clinical Sciences, College of Veterinary Medicine, University of Tennessee, 2407 River Drive, Knoxville, TN 37996, USA

E-mail address: draditic@utk.edu

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Complementary approaches can include herbs, supplements, acupuncture, and other modalities that are rational and supported by evidence to alleviate physical symptoms, improve quality of life, and also prevent diseases in veterinary patients. There is a growing consumer demand for an integrative health care approach, and veterinarians are being challenged to know more about these nontraditional therapies.^{2,3} In recognition and response to these demands, the American Veterinary Medicine Association admitted the American Veterinary Holistic Medical Association and the American Academy of Veterinary Acupuncture into the House of Delegates as constituent allied veterinary organizations.^{4,5}

URINARY TRACT INFECTIONS

Veterinarians frequently manage patients with simple lower urinary tract diseases (LUTD), such as bacterial urinary tract infections (UTIs), with conventional therapies with good outcomes. Most simple infections resolve with an approximate 2-week course of oral antimicrobials, but persistent or recurrent UTIs (RUTIs) that involve refractory bacterial isolates can be difficult to treat using conventional antimicrobial therapy alone.⁶ UTIs develop when there is a breach in host defense mechanisms, which allows virulent microbes to adhere, multiply, and persist within the urinary tract. Bacterial UTIs may affect 14% of all dogs during their lifetimes, and they are more common in females. Although less common in cats, bacterial UTIs usually occur in those older than 10 years, with incidence increasing with age. In these patients, repeated courses of antimicrobials are typically ineffective at achieving long-term bladder sterility and a nonantimicrobial preventative strategy is needed.⁷ The bacteria that most commonly cause UTIs are similar in dogs and cats; *Escherichia coli* infections account for more than one-half of all positive urine cultures.^{6,7}

In the human literature, the efficacy of different forms of nonantimicrobial prophylaxis are being assessed for effectiveness and tolerability in adults with recurrent UTIs. UTIs are more common in women, accounting for nearly 25% of all infections, with around 50% to 60% developing UTIs in their lifetimes. Similar to veterinary patients it has been shown that *E coli* is the organism that causes UTIs in most human patients, and RUTIs are reinfection by the same pathogen.⁸ In human medicine managing, RUTI is primarily done with repeated or continuous antimicrobial therapy, and resistance in common urinary pathogens is increasing because of the overuse and misuse of antimicrobials. It has been shown that resistance patterns vary by geographic location but are rising nationwide and globally.⁹

Because of the growing concern for antimicrobial resistance, nonantimicrobial, unconventional treatments evaluated in the human literature include vaginal vaccines, behavioral modifications, vaginal and oral estrogens, oral immunostimulant *E coli* fractions OM-39 (Uro-Vaxom), cranberry supplements, oral and vaginal probiotics, herbs or herbal preparations, mannose, and acupuncture.^{8–13}

A systematic review and meta-analysis of randomized controlled trials (RCTs) in human research included 17 studies with data for 2165 patients, and the Jadad score was used to assess the risk of bias (0–2, high risks and 3–5, low risks). The Jadad scale assigns scores for the best-quality trial from 0 to 5 based on the following: (1) the study is randomized; (2) the intervention is double blind; (3) study withdrawals are accounted for and described; (4) the randomization procedure is adequately performed using an appropriate method; and (5) the blinding is adequately performed using placebo. An assessment of effectiveness, tolerability, and safety of nonantibiotic prophylaxis in adults with recurrent UTIs was reported. The oral immunostimulant OM-89 (Uro-Vaxom), an extract of 18 different serotypes of heat-killed uropathogenic *E coli*,

decreased the rate of UTI recurrence (Jadad score 3, relative risk [RR] 0.61, 95% confidence interval [CI] 0.48–0.78). A vaginal vaccine Urovac was shown to reduce UTI recurrence (Jadad score 3, RR 0.81, 95% CI 0.68–0.96), whereas vaginal estrogens demonstrated a trend toward preventing UTI recurrence (Jadad score 2.5, RR 0.42, 95% CI 0.16–1.10). Cranberries and acupuncture were reported to decrease recurrent UTIs (Jadad score 4, RR 0.53, 95% CI 0.33–0.83) and (Jadad score 2, RR 0.48, 95% CI 0.29–0.79) respectively, whereas oral estrogens and lactobacilli prophylaxis did not decrease the rate of UTI recurrence.¹²

The evidence base for the use of relevant and practical nonantimicrobial treatments of UTIs in veterinary medicine was assessed via a literature search in dog/cat, in vitro using dog/cat cells, in vivo other species especially humans, and in vitro cell lines. From this review of the current literature, consideration of common nonantimicrobial treatments that could be relevant, safe, and practical for UTI in dogs and cats included cranberry supplements, mannose, oral probiotics, acupuncture, herbs, or herbal preparations.

Cranberry (Vaccinium Macrocarpon)

Cranberries have been used in the prevention of UTIs for many years. The mechanism of action has not been elucidated completely, but cranberries contain an A type proanthocyanidins (PAC), which may inhibit the adherence of P fimbriae of *E coli* to uroepithelial cells. There is no meta-analysis or RCTs evaluating the use of cranberry (*Vaccinium* species) juice or extracts and UTIs in dogs or cats. A small study using the mannose-resistant hemagglutination urine assay was used for detecting antiadhesion activity both in vitro and ex vivo following ingestion of a cranberry product in 6 beagle dogs. After one Cranranidin (Nutramax Laboratories, Inc, Lancaster, SC) tablet containing 16 mg of bioactive PACs (mean dose 1.14 mg/kg body weight) was administered daily for 3 weeks, collected urine samples were tested in an in vitro assay for the ability of P-fimbriated *E coli* to agglutinate human red blood cells. Control urine demonstrated hemagglutination activity. Antiadhesive activity appeared within 3 hours of Cranranidin administration, peaked on day 7 at $80 \pm 26.5\%$ and remained elevated on day 21 at $73 \pm 28.5\%$ over the 24-hour period. The investigators suggested that this study demonstrates metabolites of the extract in the urine of 6 healthy dogs is sufficient to reduce adhesion of Pf *E coli* and may be beneficial in preventing *E coli*-associated UTIs in dogs. Reported is the need for further studies of this extract in a clinical trial of affected dogs.¹⁴

Numerous studies in humans report that UTIs are one of the most commonly treated outpatient diseases, and complications from persistent or repeated infections result in more than a million hospital admissions annually in the United States. A 2012 updated review of 24 RCTs with 4473 participants (using cranberry juice, concentrate, tablets, or capsules to treat or prevent UTIs compared with placebo, no treatment, water, methenamine hippurate, antibiotics, or lactobacillus) was published. Compared with placebo, water, or no treatment, cranberry products did not significantly reduce the occurrence of symptomatic UTIs overall (RR 0.86, 95% CI 0.71–1.04). No significant difference in adverse effects from cranberry products compared with those of placebo/no treatment (RR 0.83, 95% CI 0.31–2.27) was noted. It was noted that the cranberry products may not have had enough potency to be effective, and the concentration of the active ingredient or type A PAC was often unknown in these studies.¹⁵

In a double-blind, double-dummy noninferiority trial, 221 premenopausal women with RUTIs were randomized to 12-month prophylaxis use of trimethoprim-sulfamethoxazole (TMP-SMX) 480 mg once daily or cranberry capsules 500 mg twice

daily. Unlike most studies, 1 capsule with 500 mg cranberry extract (Cran-Max) was used; the amount of type A PACs in the cranberry extract was 9.1 mg/g, quantified using a high-performance liquid chromatography (HPLC) method coupled with fluorescence and mass detection. The proportion of patients with at least 1 symptomatic UTI was higher in the cranberry group than in the TMP-SMX group (78.2% vs 71.1%), and median time to first recurrence was 8 months for the TMP-SMX and 4 months for the cranberry group ($P = .03$, log-rank test). The study also compared the effects of 2 different forms of prophylaxis, TMP-SMX and cranberry prophylaxis, on antimicrobial resistance among indigenous and uropathogenic *E coli* (UPEC) isolates; high resistance rates occurred after 1 month of TMP-SMX prophylaxis use. An increase in resistance to amoxicillin and the quinolones was demonstrated during use of TMP-SMX prophylaxis. A high percentage (95%) of TMP-SMX-resistant microorganisms in the feces and urine of TMP or TMP-SMX recipients was reported after only 2 weeks. In summary, in premenopausal women with RUTIs, TMP-SMX, 480 mg once daily, is more effective than cranberry capsules, 500 mg twice daily, for the prevention of RUTIs; but it should be weighed against the greater development of antimicrobial resistance.¹⁶

In vitro studies have identified the role of PAC with A-type linkages in cranberries. These studies show specific PAC inhibits P-fimbriae synthesis and induces a bacterial deformation, on both antimicrobial-susceptible and antibiotic-resistant UPEC. It has been suggested that cranberry PAC prevent bacteria from adhering to the uroepithelium of the bladder, therefore, blocking the ability of *E coli* to infect the urinary mucosa.^{17–19}

An in vitro study of antiadhesion activity of A2-linked proanthocyanidins from cranberry on UPEC and *Proteus mirabilis* strains and their possible influence on urease activity of the latter reported a significant reduction of UPEC adhesion (up to 75%) on the HT1376 cell line versus control. A reduction of adhesion (up to 75%) compared with controls as well as a reduction in motility and urease activity in the strains of *Proteus* was demonstrated. These results suggest that A2-type cranberry proanthocyanidins could aid in maintaining urinary tract health.²⁰

In summary, studies evaluating cranberry juice, capsules, and extract suggest a role in the prevention of UTIs in susceptible populations but may not be as effective in RUTIs. Standardization of A2-type proanthocyanidins is needed to perform double-blind, placebo-controlled clinical trials that can determine the role of this nonantibiotic therapy for UTIs and RUTIs as well as appropriate dosing.^{13,21,22}

Mannose

E coli is the major causative agent of UTIs and the most nonepidemic bacterial infection in humans and domestic animals. UTIs are primarily caused by UPEC (70%–95% of cases) expressing type 1 pili. At the tip of these pili, the lectin fimbriae H (FimH) is located; it enables the bacteria to adhere to oligomannosides of the glycoprotein uroplakin Ia, which is located on the uroepithelium. This initial adhesion is required for infection because it prevents the rapid clearance from the urinary tract by the bulk flow of urine enabling the invasion of the host cells.^{23,24}

The interaction of the lectin FimH, at the tip of bacterial pili, with mannose structures is critical for the ability of UPEC to colonize and invade the bladder. Synthesis and in vitro/in vivo evaluation of R-D-mannosides with the ability to block this bacteria/host cell interaction, described as *FimH antagonists*, have been reported. Initially, FimH antagonists were directly instilled into the bladder concomitantly with uropathogenic; but more recent studies in mice have evaluated nanomolar FimH antagonists as α -D-mannosides exhibiting appropriate pharmacokinetic properties for intravenous and oral treatment of UTIs.²⁵

More recent *in vivo* mice studies indicate the potential of orally available FimH antagonists for the prevention and treatment of UTIs, with a higher therapeutic effect compared with antimicrobials. Furthermore, antagonists applied together with antimicrobials were reported to have a synergistic effect on treatment outcomes. In the UTI mouse model, treatment with FimH antagonists prevented invasion of UPECs into bladder cells leading to a reduction of biofilm formation. The oral α -D-mannopyranosides tested in the UTI mouse model, including biphenyl α -D-mannopyranoside, diamidobiphenyl α -D-mannopyranoside, and monoamidobiphenyl α -D-mannopyranoside, are promising; but they still present challenging pharmacokinetics, such as low solubility and short exposure in plasma and urine, and require further study.²⁶

The reported FimH antagonists are α -D-mannosides and act as potential ligands of mannose receptors of the human host system; therefore, target selectivity of these FimH antagonists is a safety concern. Mammalian mannose receptors are present on many tissues throughout the whole body and are involved in cell-cell adhesion and serum glycoprotein homeostasis, and they intervene in the innate and the adaptive immune response by recognizing molecular patterns on pathogens. A study investigated the selectivity range of 5 FimH antagonists belonging to different compound families by comparing their affinities for FimH and 8 human mannose receptors. From this study using cell-free competitive binding assay, no adverse side effects from nonselective binding to human receptors were reported. It was concluded that FimH antagonists should be further considered as a nonantimicrobial therapy for the treatment of UTIs.²⁴

D-mannose powder is available and has been used to treat UTIs in horses, cats, and dogs, although there are no RCTs or case studies published or validated. It has been shown that *in vitro* D-mannose applied locally reduces the adherence of *E coli*, *Pseudomonas aeruginosa*, and *Streptococcus zooepidemicus* to endometrial epithelial cells in mares. It is marketed for UTI prevention in human beings as a food supplement. In 2012, a clinical trial that included 308 women greater than 18 years of age with acute UTI and a history of recurrent UTIs was reported. Patients were initially treated with ciprofloxacin 500 mg twice daily for 1 week for their UTI and then randomly allocated to one of 3 equal groups: prophylaxis with 2 g of D-mannose powder daily, 50 mg of nitrofurantoin once a day, or no prophylaxis. During the 6-month study period, there was 32% RUTIs, with the rate of RUTI significantly higher in the no-prophylaxis group (60%) compared with the groups receiving D-mannose (15%) and nitrofurantoin (20%). The risk of recurrent UTI episodes was significantly higher in the no-prophylaxis group compared with the groups that received active prophylaxis (RR 0.24 and 0.34). Diarrhea (8%) was the only adverse effect noted with the D-mannose, but it did not require discontinuation of the supplement. This study reported no difference between patients taking nitrofurantoin or D-mannose. Another study in humans with a history of uroliths and UTI evaluated a D-mannose-containing supplement and determined the supplement was well tolerated, but more RCTs and pharmacokinetics are needed to ensure efficacy and safety.^{27,28}

Studies of proximal tubular reabsorption in dogs demonstrated a separate binding site for D-mannose from D-glucose, D-deoxyglucose, and D-galactose.²⁹ Studies of mannose metabolism have been evaluated in other mammals as well as tissue distribution in rats demonstrating rapid uptake with oral administration, incorporation into plasma glycoproteins and organ glycoprotein, and are referenced in FimH antagonist research.³⁰

The author has used D-mannose in select cases of RUTIs in dogs and cats with no adverse effects and positive outcomes. Using the same laboratory for all sampling, pathogens were cultured and found resistant to an antimicrobial panel; a specific D-mannose supplement was given orally; and follow-up culture and sensitivity of urine

was performed. It was noted that the same bacterium was present but with different antimicrobial sensitivity pattern. Appropriate antimicrobial therapy was initiated with continuation of the D-mannose supplement, and the infection was eliminated as demonstrated by follow-up negative urine cultures.

Probiotics

The use of probiotics for disease treatment and prevention is growing rapidly as research explores the role of normal microbial flora and the gastrointestinal tract's role in host immune function. Data regarding probiotic use for the prevention or treatment of lower urinary tract diseases in cats and dogs is lacking. Research has primarily examined the role of probiotics in gastrointestinal diseases, with a recent small study comparing probiotic with traditional therapy for inflammatory bowel disease in dogs.^{31,32}

Proposed mechanisms in the human literature for prevention of urogenital infection by probiotics include modulating host immunity, preventing adherence of pathogenic organisms to the urogenital epithelium, and modulating growth and/or the colonization of pathogens.⁹ In woman, vaginal colonization with lactic acid-producing bacteria (LAB) is associated with a reduction in RUTIs. The role of LAB vaginal colonization was evaluated in 35 healthy spayed female dogs with no history of UTI. In this study dogs, were given an oral probiotic containing *Lactobacillus*, *Bifidobacterium*, and *Bacillus* species once daily for either 14 days or 28 days in a prospective, controlled study. A vaginal tract culture was obtained from each dog before and after oral probiotic administration. Twenty-three dogs received oral probiotic supplement for a period of 14 days, and 12 dogs received the oral probiotic for 28 days. LAB were isolated from 7 of the 35 dogs before probiotic administration, and only 6 of 35 dogs had LAB isolated after the treatment course. Administration of this oral probiotic for a 2- or 4-week period did not increase the prevalence of vaginal LAB in healthy dogs, and it was concluded that LAB are not common isolates from the vaginal vault of dogs.³³ A more recent study by the same investigator compared the vaginal microbiota of dogs with historical RUTIs to dogs with no history of UTIs. The vaginal microbial population was reported to be the similar in the two groups.³⁴

A systematic review of the human research included data from 5 different studies (294 patients) and reported no statistically significant difference in the risk for recurrent UTIs in patients receiving lactobacillus versus controls. In the 2013 review, when studies using ineffective strains and studies testing for safety were excluded, a marginally statistically significant decrease in recurrent UTI was found in patients on lactobacillus (RR 0.51, 95% CI 0.26 to 0.99, $P = .05$).³⁵ In a randomized trial comparing Lactobacillus and TMP-SMX in 252 postmenopausal women with recurrent UTIs, the 12 months' prophylaxis using probiotic did not meet the noninferiority criteria in the prevention of UTIs when compared with TMP-SMX. It was noted that unlike TMP-SMX, lactobacilli probiotic do not increase antimicrobial resistance, which is problematic in RUTIs. The investigators agreed that more research is needed to explore the role of probiotics in UTIs especially with growing pathogen resistance to antibiotic therapies.^{36,37}

Acupuncture

Human studies indicate that the UTIs in susceptible women treated with acupuncture were reduced by one-third compared with untreated woman and half the rate among woman treated by sham acupuncture.^{8,38} Human studies have looked at interstitial cystitis in woman with inconclusive results.³⁹⁻⁴¹ Sacral acupuncture in an acetic

acid bladder irritation rat model suggested a positive effect perhaps through inhibition of capsaicin-sensitive C-fiber activation.⁴²

UROLITHIASIS

Urolithiasis is a common and often challenging disorder in dogs and cats. Many factors contribute to saturation of urine with solutes that contribute to the formation of crystal formation, aggregation, and growth into uroliths. Urinary factors include concentration of solutes, urine volume, pH, concentration of solutes, and uroliths promoters and inhibitors.³¹ It has been reported in human studies that urolithiasis is the third most common urinary problem with an estimated risk factor of 8% to 15% in America and Europe and a reoccurrence rate of 10% to 23% per year, 50% in 5 to 10 years, and 75% in 20 years.⁴³ Because of the multifactorial contributors, variable response, and high rate of reoccurrence in both veterinary and human urolithiasis, complementary therapies and prevention are being investigated.

D,L-Methionine

A study evaluated use of the urinary acidifier (D,L-methionine) and an appropriate antimicrobial agent to dissolve spontaneously occurring, infection-induced struvite urocystoliths in dogs within 8 to 12 weeks. Conventional treatment of canine struvite uroliths involves appropriate antimicrobial therapy with a concurrent therapeutic diet formulated for struvite dissolution. Fourteen client-owned dogs with presumed infection-induced struvite urocystoliths were administered antimicrobials based on the culture and sensitivity results; antimicrobial therapy was continued during medical dissolution and for 2 weeks beyond radiographic evidence of urocystolith dissolution. The urinary acidifier D,L-methionine was administered during medical dissolution and for 2 weeks beyond radiographic evidence of urocystolith dissolution at an initial dosage of 100 mg/kg by mouth every 12 hours. Plasma biochemical analysis, urinalysis, urine culture, venous blood gas analysis, and abdominal radiography were performed every 4 weeks until radiographic evidence of dissolution or until uroliths did not change over 2 successive reevaluations at which time they were removed surgically. Three dogs dropped out of the study as the owners could not administer the acidifier. Urocystoliths dissolution occurred in 8 of 11 dogs with a median time of 2 months and range of 1 to 4 months. The average final dose of the D,L-methionine in which urocystoliths dissolution occurred ($n = 8$) was 97.3 ± 25.6 with a median of 104.2 and range of 56.8 to 125.0 mg/kg by mouth every 12 hours. In 3 dogs, urolith dissolution failed; surgery was performed; and analysis of the urocystoliths revealed calcium oxalate (CaOx) urocystoliths.⁴⁴

Probiotics

Cats and dogs are prone to oxalate uroliths similar to humans. In the human literature, hyperoxaluria complicated by urinary tract uroliths are an important clinical problem; the lifetime reoccurrence rate can be 50%. Mammals do not have the enzyme required to metabolize oxalate and depend on oxalate-degrading bacteria in the gut for this function.⁴⁵ Currently, there is no successful medical dissolution protocol published in either the human or veterinary literature; similarly uroliths are removed by physical methods, and dietary management for prevention is not always successful.^{31,46}

The interest in probiotics to reduce the CaOx urolithiasis risk is based on the link between the reduced prevalence of *Oxalobacter formigenes*, an oxalate-degrading bacterium in stone-forming populations. This oxalate-degrading bacterium has been detected by quantitative polymerase chain reaction in 25% of dogs with CaOx uroliths

compared with 50% of clinically healthy, age-, breed-, and sex-matched dogs and 75% in non-stone-forming breeds.^{31,45}

The use of oxalate-degrading bacteria as probiotics have demonstrated the potential for microbial metabolic interactions that may support oxalate degradation.⁴⁵ A study evaluating oxalate degradation in dogs and cats reported intestinal microflora play an important role in the regulation of intestinal oxalate, oxalate absorption, urinary excretion, and in possible oxalate urolith formation. The study evaluated the ability of the lactic acid bacteria (LAB) component of canine and feline feces to degrade oxalate in vitro. Oxalate degradation by individual canine-origin LAB was also evaluated, and the effects of various prebiotics on in vitro oxalate degradation by selected oxalate-degrading canine LAB were also evaluated.⁴⁷

A study evaluated oxalate degradation by *Bifidobacteria* species and *Lactobacillus* species isolated from the canine and feline gastrointestinal tract in vitro and selected strains in vivo in rats. *Bifidobacteria* species and *Lactobacillus* species that degraded oxalate in vitro were shown to survive gastric transit and were selected for further examination. In vitro degradation was detected for 11 of 18 of the *Lactobacillus* species, whereas oxalate degradation was not detected for any of the 13 *Bifidobacterium* species tested. A 4-week in vivo rat study demonstrated urinary oxalate levels were significantly reduced ($P < .05$) in rats fed *L. animalis* 5323 and *L. animalis* 223C but were unaltered when fed *L. murinus* 1222, *L. murinus* 3133, or placebo. It was reported that probiotic organisms vary widely in their capacity to degrade oxalate, and in vitro degradation does not uniformly translate to in vivo effect. The results have therapeutic implications and may influence the research of probiotics and oxalate metabolism.⁴⁸

Several human studies have assessed the role of probiotics with oxalate-degrading potential and reducing the risk of developing CaOx uroliths. Studies have mixed results and suggest that the ability of chronic probiotic ingestion to reduce urinary oxalate excretion may be primarily confined to individuals with absorptive (enteric) hyperoxaluria.⁴⁹

A recent study, using an oxalate-rich diet, induced in 20 healthy human subjects persistent dietary hyperoxaluria and increased plasma oxalate concentration. After 2 weeks of the oxalate-rich diet, the subjects were administered a LAB preparation (Oxadrop, VSL Pharmaceuticals, Rome, Italy) containing 3.6×10^{11} colony-forming unit (*Lactobacillus acidophilus*, *Bifidobacterium infantis*, *Streptococcus thermophiles*, and *Lactobacillus brevis*). There was no change in urinary oxalate excretion; therefore, it was concluded that it is unlikely that the LAB preparation decreased the intestinal oxalate concentration and/or absorption. The study reported the preparation may be altered in the future to select for LAB strains with the highest oxalate-degrading activity.⁵⁰

Another study assessed the oxalate-degrading ability of bacterium from human feces and fermented food to isolated 3 strains: *L. fermentum* AB1, *L. fermentum* TY5, and *L. salivarius* AB11 with efficient oxalate degradation. These strains were evaluated and demonstrated desirable properties for probiotic applications, such as good adherence to human colonic cell line HT-29 cells, tolerance to acid and bile, strong inhibition against pathogens, and absence of transferable antibiotic resistance. The reported strains survived well during gastrointestinal transit and reduced the coliform counts in feces with good a fecal recovery rate in the rat model; the study concluded that these isolates should be considered for prophylaxis of CaOx uroliths.⁵¹ It is evident because of the prevalence, incidence, and similar mammalian oxalate metabolism that a need for translational research in the use of probiotics and CaOx uroliths in human and veterinary medicine is needed.⁴⁵

N-3 Fatty Acids in Urolithiasis and Urinary Tract Tumors

Inflammation is characteristic of urolithiasis, interstitial cystitis, and urinary tract tumors (UTT). It has been noted that in cultures eating cold-water fish, urolithiasis is rare and has been attributed to the dietary intake of eicosapentaenoic acid (EPA) content in the diet. The N-6 fatty acid, arachidonic acid (AA), is a precursor for proinflammatory and pro-aggregatory prostaglandin E_2 (PGE_2), which influences renal calcium excretion and hypercalciuria. Both EPA and docosahexaenoic acid (DHA) are known to be precursors of less inflammatory precursors, such as PGE_3 . High EPA intake competitively inhibits conversion of AA to PGE_2 and other proinflammatory cytokines.^{46,52}

A study of 10 cats with a history of urolithiasis or with current urolithiasis were fed a dry food with 0.65% total omega 3 fatty acids on a dry matter basis (DMB) and an N-6 to N-3 ratio or a top-selling over-the-counter diet for 4 weeks. Blood samples were analyzed using HPLC demonstrating that EPA and DHA from the fish oil sources were readily absorbed by the cats and appear in the serum at a ratio similar to the diet fed; therefore, there is therapeutic potential of dietary EPA and DHA in cats as well as dogs.^{46,53}

A human study investigated the effects of an N-6 fatty acid, γ -linolenic acid (γ -LA), in the form of evening primrose oil (EPO) in 2 ethnic groups. Eight healthy white and 8 healthy black men were given a single 1000-mg capsule of EPO for 20 days. During the study baseline, day 10, day 20, and day 24 (4 days after cessation of EPO), 24-hour urine collections were obtained and analyzed. Calcium excretion decreased significantly in both groups by day 20, and citrate excretion increased and was statistically significant at day 20 for the black men and day 10 in white men. A statically significant decrease in the CaOx stone formation Tiselius risk index occurred in both groups. AA is released by the key enzyme phospholipase A2, which increases anion-carried phosphorylation and potentially calcium excretion. It was speculated in this study that EPO may modulate the action of fatty acid desaturases, inhibit production of AA and PGE_2 , thereby decreasing calcium excretion.⁵² In a more recent study of N-3 polyunsaturated fatty acid (PUFA), 15 health human subjects were given 30 days of a 900-mg EPA and 600-mg DHA capsule once a day. Relative supersaturation of CaOx decreased 23% from a mean $\pm$ SD of 2.01 ± 1.26 to 1.55 ± 0.83 because of a significant decrease in oxalate excretion ($P = .023$). It was concluded that CaOx formers may benefit from N-3 fatty acid supplementation, although more studies are needed.⁵⁴

Abnormal PUFA metabolism may occur in humans, which is not only related to the complex metabolic syndrome, such as diabetes and obesity, but also to urolithiasis and UTIs. In human patients with urolithiasis, an anomalous phospholipid metabolism of N-6 PUFA was recorded; in plasma and the erythrocyte phospholipid membrane of stone subjects, the observed AA/LA (N-6 to N-3) acid ratio was increased. Phospholipase A2 increases the erythrocyte anion-carrier protein phosphorylation and the oxalate exchange, revealing that idiopathic nephrolithiasis is related to abnormal phospholipid-AA metabolism. Fish oil supplementation was shown to lower calcium and oxalate urinary excretion and normalize the erythrocyte oxalate exchange.⁵⁵

There is cumulative evidence in the literature linking the low-grade chronic inflammatory state of the urinary tract mucosa, calculi, and UTTs. UTTs are the 10th cause of human cancer worldwide; similar to veterinary medicine, direct mortality is not high, but morbidity and recurrence of these tumors is challenging. In human UTTs, the death rate caused by bladder neoplasia has not diminished over decades; we have also not seen significant advancement in veterinary medicine with bladder neoplasia. Evidence strongly suggests that moderate, long-standing consumption of dietary supplementation of N-3 PUFAs may improve the chances of avoiding recurrent CaOx uroliths as well as the development of UTT.⁵⁵

Acrylamide (ACR) is a vinyl monomer that improves the aqueous solubility, adhesion, and cross-linking of polymers. The International Agency for Research on Cancer classified ACR in 1994 as probably carcinogenic to humans (group 2A) from sufficient evidence for carcinogenicity in experimental animals and concluded that current ACR levels in food items may indicate a human health concern. A study investigated whether various dietary fats affected preneoplastic lesions of the urinary tract in ACR-treated mice.

Eighty Kunming mice were initiated with ACR at dose 10 mg/kg and fed with 6% or 10% of the following PUFA: fish oil (enriched N-3), corn oil (enriched N-6), olein (enriched N-9 or essential fatty acids [EFA] deficiency inducer), and a commercial chow. After 22 weeks the mice were euthanized and macroscopic and biochemical evaluation of liver showed fatty acid tissue profiles correlated closely with dietary source. In the ACR mice with olein, both macroscopic and biochemical evaluations demonstrated EFA-deficient (EFAD) characteristics. The corn oil (N-6) and olein (EFAD) diet had a significant promoting effect on urothelial lesions induced by ACR, with a greater number of mice developing simple hyperplasia without atypia or dysplasia/carcinoma in situ lesions, in comparison with fish oil and commercial dietary groups. It was concluded longer investigations are necessary to induce full-blown invasive carcinomas in the ACR mice model as well as to study the role of different dietary PUFA.⁵⁶

As naturally occurring canine transitional cell carcinoma (TCC) of the urinary bladder is very similar to human invasive bladder cancer, spontaneous canine TCC has been applied as a relevant animal model of humans in a translational study using metabolomics to identify early makers for screening and monitoring treatments. Metabolomics is an emerging research used to analyze a large number of small molecules in a single step and promises immense potential for discovering metabolite markers for disease detection, monitoring, and evaluating treatments. In this study, nuclear magnetic resonance (NMR) combined with statistical analysis was used to profile and compare the urine metabolites from dogs with TCC and healthy control dogs with no bladder disease; the approach was proven to be a powerful method for TCC biomarker discovery. Six individual metabolites (urea, choline, methylguanidine, citrate, acetone, and β -hydroxybutyrate) between samples from dogs with TCC and the control group based on the Student t-test were selected to build a separate partial least square-discriminant analysis model. The results showed good classification between TCC and control groups; the sensitivity and specificity of the model were 86% and 78%, respectively. In the translational model, it was concluded that urine metabolic profiling may have potential for early detection of bladder cancer and of bladder cancer recurrence following treatment and may enhance our understanding of the mechanisms involved.⁵⁷

HERBS AND LOWER URINARY TRACT DISEASES

Herbal medicines are in great demand in both developed and developing countries because of their wide biological activities and lesser costs. With Internet and increased emphasis on a global economy, consumers have a greater access to herbal products from anywhere in the world. There is much research evaluating the use of herbs and to develop new conventional drugs by isolating bioactive compounds in herbs. Although benefits can be derived from herbs alone, the areas of concern include product contamination and/or adulteration, potential toxicity, and herb/drug interactions. In many countries, including the United States, herbal medicines are not regulated as extensively; the 1994 Dietary Supplement Act classifies herbal products as dietary supplements. Other countries attempt to prove efficacy and safety of

herbal medicines are superior to the United States. The Commission E monographs in Germany are an example combining the scientific, health care, and industry leaders producing guidelines grounded in evidence-based knowledge. Prescribing of herbal medicines among health care providers in Germany is common. With the cost implications of a worldwide herbal medicine market estimated at \$83 billion, implementation of standards for growing, selecting, manufacturing, conducting appropriate clinical trials and treating patients with herbal medicines is necessary. The World Health Organization (WHO) has provided technical guidelines to standardize herbal medicines.⁵⁸

With an increasing popularity of herbal remedies by consumers, research has advanced to develop analytical techniques, as there is a need to identify and maintain uniform quality, quantification, and standardize these products. Advanced analysis techniques to characterize chemical structure, bioactivity, quality control, safety, and efficacy are available and include mass spectrometry (MS), NMR, HPLC, gas chromatography-MS, capillary electrophoresis, liquid chromatography beam MS, fluorescence detector, and flame ionization detector. These analytical techniques, although not recognized by pertinent authorities (the Food and Drug Administration), have identified many potential bioactive substances, including glycosides, saponins, phenolic compounds, flavones, quinones, xanthenes, alkaloids, benzopyrone derivatives, and terpenes in foods, plants, and herbal preparations.^{59–62}

It is known that more than 80% of drug substances are either directly derived from natural products or developed from natural compounds, such as antimicrobial, cardiovascular, immunosuppressive, and anticancer drugs. Active ingredients in herbal extracts for potential drug targets are identified by using comparative gene profiling studies. Metabolomics studies can also be used to screen herbs and identify mechanisms of action. Bioactive compound from transgenic plants or microbes for therapeutic application reduces the cost of drug development by applying genetic engineering.^{63–66}

It should be noted, despite advanced techniques for drug discovery, understanding plant/herbs has formed the basis of traditional medicine systems, which have existed and been used for thousands of years. For example, the first drug-screening experiments of the antimalarial activity of *Artemisia annua* extracts were poor, and scientists ignored the results. Later cold-water extracts demonstrated complete inhibition of mouse malaria as Artemisinin, a novel antimalarial drug that is heat labile, was finally discovered in China. Knowledge accumulated in traditional medicine plays an important role in increasing the success rate of drug discovery. It has been estimated that the success rate of a synthetic route for developing new drugs may be 1 per 10,000, whereas the success rate based on medicinal plants used in traditional medicinal systems can be as high as one-quarter or more.⁶³

Plants can have antimicrobial, antiinflammatory, and diuretic properties, which may be beneficial in lower urinary tract diseases. These important properties of plants provide a role in the treatment and/or prevention of lower urinary diseases, such as UTIs, RUTIs, and urolithiasis. Currently the WHO reports about 80% of the world population relies on plant-based medicine; there is a move toward the use of natural products for the prevention and treatment of diseases in humans and animals in developed countries.^{59,67}

Herbs and Urinary Tract Infections

Herbs and herbal formulas have been used in traditional medicine for the treatment of lower urinary tract diseases in many cultures, but there is little research in dog or cat lower urinary diseases. Studies of the Japanese formula, choreito was evaluated

in vitro and in vivo in cat and cat urine. An in vivo study reported improvement of struvite-related lower urinary tract signs in 12 cats fed the herbal formula.^{68–70} Another study evaluated 3 commonly used herbal treatments recommended for use in cats with lower urinary tract disease (LUTD) including 2 formulas (1) San Ren Tang (SRT) and (2) Wei Ling Tang (WLT) and the single herb (3) Alisma (A). It was hypothesized that these 3 herbal preparations would induce increased urine volume and decreased urine saturation for CaOx and struvite. In this study, 6 healthy, spayed female, adult cats were evaluated in a placebo-controlled, randomized, crossover design study randomized to 1 of 4 treatments, including placebo, SRT, WLT, or A. Treatment was for 2 weeks each with a 1-week washout period between treatments. At the end of each treatment period, a 24-hour urine sample was collected with urine volume and urine biochemistries measured. Urine saturation for struvite and CaOx was estimated using EQUIL 1.5b (College of Medicine, The University of Florida, Gainesville, FL). Analysis of variance was used to analyze data statistically if distributed normally, and Kruskal-Wallis was used to analyze data statistically if data were not distributed normally. No differences were found in 24-hour urinary analyte excretions, 24-hour urine volume, urine pH, or 24-hour urinary saturation for CaOx or struvite between treatments. Results did not support the use of SRT, WLT, or A for increasing urine volume or for decreasing urine saturation for CaOx or struvite in cats; but no adverse events occurred during the study; therefore, these herbal compounds seem safe in cats.⁷¹

There are published studies in vivo in humans and rats and in vitro evaluating the antibacterial activity of selective plants and herbal formulas. Mechanisms of action include bioactive compounds in these plants having antiadhesive properties or direct cytotoxic activity. An in vitro study of 20 herb extracts against UPEC and T24 human bladder carcinoma cells reported that none of the herbs demonstrated direct cytotoxic effects. Two extracts prepared from *Agropyron repens* L and *Zea mays* decreased bacterial adhesion by interacting with bacterial outer membrane proteins. Three herb extracts, *Betula* spp, *Orthosiphon stamineus*, and *Urtica* spp, showed antiadhesive effects by interacting with T24 cell surface.⁷²

Arctostaphylos uva ursi folium (bearberry leaf) has traditionally been used for medicinal purposes in Europe and America for the treatment of UTIs and may have direct antibacterial effects. The bioactive substance arbutin is rapidly absorbed and undergoes hepatic conjugation to hydroquinone (HQ) conjugates. The proposed mechanism of action is via HQ conjugates carried to the kidneys for excretion followed by decomposition, releasing the HQ to act as a direct antimicrobial agent.^{73,74} Women (n = 57) with recurrent cystitis volunteered for a double-blind trial and were randomized to take placebo or an uva ursi extract for 1 month. Women in the uva ursi group had no episodes of cystitis in the following year compared with 23% of women who took placebo. No adverse effects were reported in this study; but there have been concerns as free HQ used in cosmetic products has nephrotoxic, hepatotoxic, or carcinogenic potential.⁷³

A recent review of HQ toxicity noted arbutin metabolism forms primarily HQ conjugates, HQ glucuronide and sulfate, which are nontoxic and have rapid excretion via the urine. It is concluded that extracts of uva ursi folium are a safe therapeutic option for the treatment of lower UTIs, and no human case reports have been published regarding toxicity of this herb. Herbal medicinal preparations of bearberry leaves have been standardized or quantified to the content of arbutin and is one of the herbs monographed in Germany's Commission E.^{75,76}

In a current study of *Aframomum melegueta*, bacteriocidal activity was evaluated against *E coli*, *L monocytogenes*, methicillin-resistant *Staphylococcus aureus*, and *S aureus*. Using NMR and MS, diterpenes were isolated and tested for their

antibacterial effects. Two of the diterpenes exhibited more potent antibacterial activity compared with current clinically used antibiotics.⁷⁷ Other plant extracts have been evaluated for use in UTIs with the bioactive compound berberines, including *Mahonia aquifolium* (Oregon grape), *Hydrastis canadensis* (goldenseal), and *Coptis chinensis* (goldthread).⁷³

Other plant extracts have been evaluated for their activity against uropathogens with elucidation of the bioactive compound and/or mechanism of action. P and type 1 fimbriae play a particular role in the adhesiveness of UPECs and bacterial binding is also mediated by the hydrophobic interactions between uropathogenic rods and uroepithelial cell surfaces. Correlation of bacterial adherence and increasing cell surface hydrophobicity of the microorganism has been demonstrated as another mechanism for infection. Studies of herbs and their metabolites interfering with UPECs by preventing bacterial adherence, increasing hydrophobicity, and development of biofilm rather than direct cytotoxic mechanisms are considered in vitro studies.^{78–83}

Mechanisms of resistance is important in the development of integrative strategies for resolving RUTIs. Active efflux of antibiotics is one of the major mechanisms of drug resistance in bacteria. Multidrug-resistant (MDR) pathogens express and overproduce efflux pumps that expel and extrude of the antibiotics from inside the cell. Combining efflux-pump inhibitors (EPIs) together with antibiotics can reduce the invasiveness of these drug-resistant pathogens. In vitro studies have demonstrated herbs and/or metabolites that may be effective contain EPI properties such as seen with hydroalcoholic extract of *Terminalia chebula* fruits, which exhibited strong antibacterial activity against MDR uropathogens.^{79,80}

Herbs and Urolithiasis

Diet manipulation to influence the volume, pH, and solute concentration has been the conventional treatment of urolithiasis in veterinary medicine. Besides the in vivo and in vitro studies discussed previously, herbal medicine has not been evaluated by RCTs in veterinary urolithiasis. A published case series between 2001 and 2009 identified 33 dogs and 13 cats with radiographic or ultrasonographic evidence of urolithiasis that were treated with Chinese herbal supplement, CrystaClair, a proprietary blend⁸⁴ developed by the investigators. Patients were administered CrystaClair at a dosage 0.5 g per 15 lb body weight twice a day and were monitored with abdominal radiographs to assess uroliths number and size on presentation and periodically during herbal treatment until uroliths dissolution or treatment was discontinued. No adverse effects were reported, and the overall duration of treatment and end point was defined as uroliths resolution or herbal medicine discontinued. Dissolution was reported in 56.5% of patients (58% of dogs and 54% of cats) over 4 to 60 weeks with CrystaClair. The investigators stated this case series was not ideal with unknown uroliths but that this was similar to clinical practice whereby uroliths composition is often unknown. Problems with study design, definition of known prior (surgery and crystalluria), and the use of diet therapy make it difficult to interpret this study.⁸⁵

Fourteen Kampo extracts (10 mg/mL) were examined in vitro by assessing whether they could inhibit 2 critical steps in the early process of CaOx stone formation, namely, crystal aggregation and adhesion to the renal tubular epithelium of Madin-Darby canine kidney cells. The inhibitory effect of the extracts on stone formation was examined by using an ethylene glycol (EG) rat model. Sanshishi (*Gardeniae Fructus*) and Takusha (*Alismatis Rhizoma*) inhibited CaOx monohydrate crystal aggregation (84.5% and 64.2%, respectively) and crystal adhesion to Madin-Darby canine kidney cells (88.2% and 54.6%, respectively). The combination of these

two herbs in a Kampo formula, known as Gorin-san, showed significantly stronger inhibitory activities; the study concluded Gorin-san may be a prophylaxis for CaOx urolithiasis.⁸⁶

It is estimated that 12% of the world population experiences renal stone disease with a recurrence rate of 70% to 80% in men and 47% to 60% in women. The pathophysiological mechanisms of urolithiasis are still not well understood, and it is currently the third most common urologic disease after UTIs and prostatic disease in human medicine.⁸⁷ In traditional human medicine systems, herbs have been used for thousands of years to manage urolithiasis especially CaOx. Herbs and herbal preparations have been evaluated for their antiurolithic properties in vivo in humans and in the EG-induced nephrocalcinosis rat model. A traditional Chinese formula, Wu ling san (WLS), using the EG-induced nephrocalcinosis rat model used 41 male Sprague-Dawley (SD) rats divided into 4 groups: normal, placebo (EG and starch), low-dose 375 mg/kg of WLS and EG, and high-dose WLS 1.125 mg/kg and EG. Biochemical data of urine and serum were collected; at the end of 4 weeks, the rats were euthanized and renal tissues examined by polarized light microscopy and the crystal deposits evaluated by a semiquantitative scoring method using computer software (*ImageScoring*). In summary, WLS inhibited the deposition of CaOx crystal and lowered the incidence of stones in rats ($P = .035$) as well as reduced the severity of CaOx crystal deposits in rat kidneys demonstrating antiurolithic properties.⁸⁸

A prospective RCT to investigate the effect of WLS in recurrent CaOx nephrolithiasis human patients reported urine output increased after treatment with WLS group, and this increase was significant when compared with the change in urine output in patients treated with placebo without causing electrolyte imbalance. The study limitations of this study are the small number of subjects (28) and that urine oxalate and citrate levels were not measured. Another limitation was using the daily amount of urine output as a marker of drug efficacy, as the environment temperature, the water content of meal, and the amount of insensible water losses will affect urine output.⁸⁹ A study evaluated WSL and doxorubicin hydrochloride (Adriamycin [ADR]) nephrotoxicity in rats demonstrating that WSL ameliorates ADR-induced nephrotic syndrome in rats. The results suggested that WSL ameliorates ADR-induced proteinuria and podocyte injury, whereas gene analysis demonstrated a suppression of renal overexpression of nephrin mRNA and protein by WLS. Radioimmunoassay showed that WLS suppressed ADR-induced increased renal angiotensin II content in rats. Some of these nephron-protective mechanisms of WSL may be involved in its antiurolithic mechanisms.⁹⁰

There are other proprietary herbal formulas for the treatment and prevention of urolithiasis, including Cystone.⁹¹ Recurrent kidney stone formers with documented CaOx stones were recruited into a 2-phased study to assess safety and effectiveness of Cystone, an herbal treatment used for the prevention of kidney stones. In the first phase, a randomized double-blinded 12-week crossover study assessed the effect of Cystone versus placebo on urinary supersaturation, whereas the second phase was an open-label 1-year study of Cystone to determine if renal stone burden decreased, as assessed by quantitative and subjective assessment of computed tomography. The results revealed no statistically significant effect of Cystone on short (6 weeks) or long (52 weeks) urinary composition, and average renal stone burden increased rather than decreased. This study did not support the efficacy of the herbal formula Cystone to treat human CaOx stone formers.⁹² An in vitro crystallization study of 5 herbs (*Folium pyrosiae*, *Desmodium styracifolium*, *Phyllanthus niruri*, *Orthosiphon stamineus*, and Cystone) in synthetic urine concluded all extracts decreased the rate of crystal growth, but Cystone had the greatest impact on all risk factors measured.⁹³

The single herbs *Poria cocos* and *Alisma orientalis* found in many traditional herbal formulas have been evaluated with numerous rat in vivo studies to determine the bioactive compounds and mechanism of action as diuretic, antiurolithic, and renal protective. These herbs contain tetracyclic triterpenes and steroids, which have a similar structure to aldosterone nucleus structure and may exert diuretic effect through competitive inhibition of aldosterone receptors in different parts of tubular reabsorption to increase urine output.⁹⁴ Studies of different fractions of *Poria cocos* administered to rats all demonstrate its diuretic effects while trying to elucidate its mechanism of action. One study suggested at least 2 mechanisms for its observed diuretic activity with the potassium ion-saving diuretic effect related to the triterpenoid components. A metabonomic study of *Poria* reported by partial least squares-discriminate analysis 15 biomarkers in rat urine were identified and 11 of them were related to the pathway of adenine metabolism and amino acid metabolism.⁹⁵

Extracts of *Alismatis rhizoma* (AR) or *Alisma orientale* ("Zexie" in Chinese) are well-known traditional Chinese medicines and are used as agents for diuresis. It has been reported using the EG rat model for urolithiasis that similar to *Poria cocos*, total triterpenoids are the antiurolithiatic active constituents of AR.⁹⁶ An in vivo study evaluated different fractions orally administered to rats measuring urinary excretion rate, pH and electrolyte excretion were measured in the urine. AR fractions demonstrated diuretic activity and inhibiting diuretic activity. The components with strong polarities in AR may have antidiuretic activities, which might be an effect of promoting the sodium-chloride cotransporter in the distal tubule.⁹⁷ In another in vivo study, the ethanol extract (EE) and the aqueous extract of AR were orally administered to rats to further elucidate the diuretic mechanisms of this herb. This study identified that EE but not AE presents a notable diuretic effect, and EE had diuretic and antidiuretic effects, which seems to be related to the sodium-chloride cotransporter in the renal distal convoluting tubule.⁹⁷ A study to investigate the subchronic toxicity of triterpene-enriched extract from AR (TEAR), as a 90-day oral toxicity study, was conducted in rats. No mortality or obvious treatment-related clinical signs, hematology, urinalysis parameters, and macroscopic or microscopic examinations were observed; it was concluded that the no-observed-adverse-effect level for TEAR was 1440 mg/kg/d in both sexes.⁹⁸

Another herbal preparation that has been evaluated is Canephron N (CAN) the main ingredients of which are centaury (*Centaureum erythraea*), lovage (*Levisticum officinale*), and rosemary (*Rosmarinus officinalis*). The active ingredients are phenolic glycosides, phenolcarboxylic acids, phthalides, secoiridoids, essential oils, and flavonoids. In the Community of Independent States, CAN is registered for prophylaxis against and after removal of uroliths as well as RUTIs, pyelonephritis, glomerulonephritis, and interstitial nephritis. Because of its diuretic, spasmolytic, antiinflammatory, antioxidative, antibacterial, and nephroprotective properties, CAN seems to have a positive clinical impact on infectious and inflammatory processes within the urinary tract. Studies indicate this herbal remedy may have a positive effect in patients with urolithiasis and after stone removal and on spontaneous elimination of small CaOx stones. It may also reduce the risk of uroliths. Adult and children dosing is reported, and no adverse events were reported in the 1762 pregnant women treated with the compound demonstrating the safety of this herbal preparation.^{99,100}

Urolithiasis is the third most common urinary problem sparing no geographic, racial, or cultural boundaries; reoccurrence is common. As herbs and herbal formulas have been used by many cultures, numerous studies using the in vivo EG antiurolithiatic rat model of CaOx have been published.^{101–118} These studies suggest that herbs could provide integrative therapy for urolithiasis treatment and prevention. Attention to the

knowledge accumulated through traditional herbal medicine will improve the development of effective therapies. Ecological ethics should be upheld to preserve biodiversity while we explore and use natural therapies and drug discovery.⁶³

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